Forms of music appreciation, social context, and recommendation source uniquely impact felt experience

Sophia K. Mehdizadeh[1, 2], *Grace Leslie*[1, 3]

[1]ATLAS Institute, University of Colorado Boulder
[2]Institute of Cognitive Science, University of Colorado Boulder
[3]College of Music, University of Colorado Boulder

Music listening experiences, while highly rewarding, are also highly individualized and context-dependent. While several frameworks have characterized multiple facets of music appreciation, few studies have compared how different types of music appreciation affect behavior. Such research is particularly relevant in the context of adolescence, a critical period for behavioral development and music engagement. Drawing from frameworks like the BMRQ (Mas-Herrero et al., 2013) and Structure-Self-Source (Thompson et al., 2022), we investigate the behavioral effects of two types of music appreciation in a late-adolescent population.

We designed two prompts to collect participant-selected favorite stimuli targeting "sensory-motor" and "identity-related" music appreciation. In a survey, participants (N = 125, ages 18-25) responded to these prompts and other trait-level inventories. Next, participants completed a deceptive shared music listening task adapted from Curzel et al. (2024) where they listened to these favorite music stimuli either alone or with simulated online participants. Participants also listened to unfamiliar, researcher-selected songs they were told were chosen by either another participant or a recommendation algorithm. After each song, we collected experience ratings (e.g., pleasure, musical connectedness) and a measure of prosocial behavior.

While both types of music appreciation were equally pleasurable ($z = 1.15$, $p > 0.05$), identity-related songs elicited higher feelings of connectedness than sensory-motor songs ($z = 13.96$, $p < 0.001$, $d = 1.25$). Unfamiliar music was rated higher in both pleasure ($z = 3.14$, $p < 0.05$, $d = 0.28$) and connectedness ($z = 4.29$, $p < 0.001$, $d = 0.38$) if it was allegedly chosen by another participant rather than a recommendation algorithm. Effects of group versus solo listening, as well as correlations between listening experiences and prosocial behavior, were small but insignificant (all $p > 0.05$).

These findings illustrate how different types of music appreciation and music discovery may contribute uniquely to felt experience during late adolescence. We demonstrate the feasibility of disentangling music appreciation in an experimental setting, extending existing methods in music cognition and providing new perspectives for studies of music reward and appreciation.